

Figure 1: A small RADOS cluster

ID	Binary Data	Metadata
1234	0101010101010100110101010010 0101010101010100110101010010 0101010101010100110101010010	name1 value1 name2 value2 nameN valueN

Figure 2: Ceph metadata

Figure 3 shows how the CRUSH algorithm strategically places objects in rows and cabinets.

Installation steps

Ceph is open source software that can be found in the Linux repositories. Given below is the step-by-step procedure to deploy a Ceph storage cluster.

The pre-requisites are:

- A Ceph node set-up
- The need to create a user on each node

```
ssh user@ceph-server
sudo useradd -d /home/ceph -m ceph
sudo passwd ceph
```

Add root privileges to the node, as follows:

```
echo "ceph ALL = (root) NOPASSWD:ALL" | sudo tee /etc/sudoers.d/ceph
sudo chmod 0440 /etc/sudoers.d/ceph
```

Install SSH server, as shown below:

```
sudo apt-get install openssh-server
sudo yum install openssh-server
```

Configure your Ceph-deploy admin node with password-less SSH access to each Ceph node. Leave the passphrase empty:

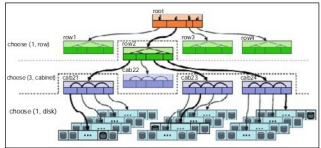


Figure 3: CRUSH algorithm strategically places objects in rows and cabinets

ssh-keygen

Generating public/private key pair.

Enter file in which to save the key (/ceph-client/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /ceph-client/.ssh/id_rsa.
Your public key has been saved in /ceph-client/.ssh/id_rsa.pub.

Copy the key to each Ceph node:

```
ssh-copy-id ceph@ceph-server
```

Modify the `~/ssh/config` file of your Ceph-deploy admin node so that it logs in to Ceph nodes as the user you created (e.g., ceph):

Host ceph-server

Hostname ceph-server.fqdn-or-ip-address.com

User ceph

Ensure connectivity using ping with host names (i.e., not IP addresses). Address the host name resolution issues and firewall issues, as necessary.

The Ceph deployment set-up

Ceph can be deployed using both kinds of package management functionality, rpm and dpkg. I am retrieving packages from the Ceph repository on an Ubuntu system, so I will be using the `apt` command to do this. RedHat Linux users need to follow the `Yum` command prompt to do the package retrieval.

Add the `release` key, as follows:

```
wget -q -O- 'https://ceph.com/git/?p=ceph.git;a=blob_plain;f=keys/release.asc' | sudo apt-key add -
```

Add the Ceph packages to your repository. Replace `{ceph-stable-release}` with a stable Ceph release (e.g., cuttlefish, dumpling, etc), as follows:

```
echo deb http://ceph.com/debian-{ceph-stable-release}/ $(lsb_release -sc)
```